

Part I

Foundations: Why Rates Are Different

Rates Aren't Prices

1.1 The habit you have to unlearn

If you have priced an equity option, you already carry a mental model that will actively mislead you here. A stock price is a level: it starts positive, and under geometric Brownian motion it can only ever multiply by a positive factor, so it stays positive. Returns are what is roughly normally distributed; the price itself is lognormal. That single fact — lognormal price, normal return — is the quiet assumption underneath Black–Scholes, and it is the assumption you have to put down before touching an interest rate.

An interest rate is not a price. It is a **spread**: the difference in value, expressed as a rate of exchange, between money now and money later. Spreads can be negative. The eurozone ran negative policy rates from 2014 to 2022; Swiss and Japanese short rates went negative earlier and stayed there for years. A model that structurally forbids negative values — as a lognormal model does, by construction, since $e^{\text{anything}} > 0$ — cannot price an option on something that occasionally goes negative, not approximately, but as a matter of the model's own algebra. Ask a lognormal model for the probability that a rate is below zero and it will answer, correctly according to its own assumptions and incorrectly according to the world, zero.

PITFALL

It is tempting to think “rates are close to zero, so lognormal vs. normal barely matters, it's a second-order modelling choice.” It is not. The entire shape of the volatility surface, the sign conventions on skew, and whether the model can even be evaluated at low-strike options all change. A trader who learned options on equities and applies equity intuition uncritically to a rates desk will misprice the wings of the smile and misread what “25 bp vol” even means. This is the single most consequential habit to unlearn before reading further.

1.2 Two ways to describe a random rate

Suppose a forward rate F is 3.50% today, and you want to describe how uncertain it is in one year's time. There are two natural ways to state that uncertainty, and the rates market uses both, side by side, every day.

Lognormal (relative) volatility, σ , states the uncertainty as a *percentage of the rate itself*. “20% lognormal vol” means the one-standard-deviation move over the horizon is roughly $\pm 20\%$ of 3.50%, i.e. about $\pm 0.70\%$, or ± 70 basis points (bp). This is the natural language for a rate that is not close to zero and is not expected to change sign: the dynamics are $dF = \sigma F dW$, so the size of a move scales with the level of F . Double the rate and, all else equal, the absolute size of a typical move doubles too.

Normal (absolute) volatility, σ_n , states the uncertainty directly in the rate's own units — basis points per year. “60 bp/yr normal vol” means the one-standard-deviation move over one year is ± 60 basis points, *full stop*, independent of whether the rate started at 0.50% or 5.00%. The dynamics are $dF = \sigma_n dW$: an absolute random walk. A normal rate can go negative with positive probability, however small, which is exactly why the market adopted normal quoting

during the low- and negative-rate years and has, in USD and most other major currencies, never fully gone back.

Key result: what a bp vol means

A normal volatility of $\sigma_n = 60$ bp/yr states that the one-standard-deviation move in the *rate itself* over one year is ± 60 basis points. It is an *absolute* statement about the rate, independent of the level of the rate. A lognormal volatility of $\sigma = 20\%$ states that the move is $\pm 20\%$ of the *current rate* — a *relative* statement whose bp-equivalent changes every time the rate moves.

Converting between them

Near the money, the two quotes are related by a simple rule that every rates desk carries in its head:

$$\sigma_n \approx \sigma \times F$$

where F is the forward rate expressed as a decimal. This is exact only in the limit of small volatility and short maturity, but it is accurate enough for a sanity check on any live quote.

Worked example 1.1 — normal-to-lognormal conversion.

A 5-year SOFR forward rate is $F = 3.65\%$, quoted with a normal volatility of $\sigma_n = 95$ bp/yr. What is the approximately equivalent lognormal volatility?

Step 1 — rearrange the approximation.

$$\sigma \approx \frac{\sigma_n}{F} = \frac{0.0095}{0.0365} = 0.2603$$

Step 2 — express as a percentage.

$$\sigma \approx 26.0\%$$

Sanity check. $26.0\% \times 3.65\% = 0.949\% = 94.9$ bp, which recovers the 95 bp normal quote to within rounding. The conversion is self-consistent.

Interpretation. If this same forward rate were 0.50% instead of 3.65% — as USD rates were for much of 2020–2021 — the same 95 bp normal vol would imply a lognormal vol of $0.0095/0.0050 = 190\%$, an absurd number that no lognormal model could sensibly use. This is the concrete reason the market moved to normal quoting when rates approached zero: the lognormal quote becomes numerically unusable exactly when the rate that defines it becomes small, which is precisely when accurate pricing matters most for hedging.

1.3 Why this is chapter one, not chapter nine

In most treatments, normal-versus-lognormal is a footnote buried deep in a chapter on volatility smiles. It belongs here instead, first, because almost every later decision in this book is a consequence of it:

- **Which pricing formula applies.** Black's formula assumes lognormal dynamics; Bachelier's formula assumes normal dynamics. Chapters 6 and 7 derive both, and you cannot read a market quote correctly without knowing which one is being used.
- **What the BGM model actually models.** The original Brace–Gatarek–Musiela construction is a *lognormal* model of forward rates, chosen specifically because it reproduces Black's

formula exactly for caplets (Chapter 11). Its later extensions to *displaced* and *shifted* diffusions (Chapter 15) exist entirely to fix the negative-rate problem this chapter just described, without discarding the lognormal machinery that makes the model tractable.

- **How a Greek is read.** A vega quoted against normal vol and a vega quoted against lognormal vol are different numbers for the same position. Mixing them up is a real, recurring error on real trading desks, not a theoretical curiosity.

ON THE DESK

On a SOFR options desk, quotes for caps, floors and swaptions are overwhelmingly given in normal (bp) vol today, a habit that hardened during the near-zero and negative-rate years in USD, EUR and JPY markets and never fully reversed even as rates rose back above 4–5% in 2022–2023. A trader hearing “70 normal on the 1y10y” converts that to a premium in their head before the sentence is finished; Chapter 7 builds the arithmetic behind that mental shortcut. When a system pulls a volatility number from a vendor feed, the single most common integration bug is treating a normal vol as if it were lognormal, or vice versa — the number silently loads without error, and every downstream price is wrong by an amount that depends on the rate level, making the bug very hard to spot from the output alone.

1.4 The shape of the book from here

Part I finishes building the ground floor: how a discount curve is constructed (Chapter 2), and how today’s multi-curve, SOFR-based market differs from the single-curve LIBOR world every older textbook was written for (Chapter 3). Part II works through the instruments themselves — swaps, caps, floors, swaptions — and prices each one with both the lognormal and normal formulas this chapter introduced. Part III tells the story of how the industry got from short-rate models to BGM, a story that is worth reading in order because every step exists to fix a genuine problem with the step before it. Part IV builds the BGM model itself, formally but without skipping a derivation. Part V is the payoff: given live cap and swaption prices, how a calibration engine turns them into model parameters, and how that engine is actually built and run.

Chapter benchmark

Before moving on, you should be able to, without notes:

1. Explain in one sentence why a model that forces its underlying to be positive cannot correctly price an option on a rate that can go negative.
2. State the difference between normal and lognormal volatility in terms of what each holds fixed — percentage of the rate, or the rate’s own units.
3. Convert a normal (bp) vol quote to an approximate lognormal vol given the forward rate, and explain why this conversion breaks down as the forward rate approaches zero.

